Plane Wave Propagation in Lossless Media

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Lecture Objective

- To derive and examine the behavior of a plane wave propagating in a lossless medium.

Helmholtz Equation for Electric Field

- From previous lecture

$$\nabla^2 \tilde{\vec{E}} - \gamma^2 \tilde{\vec{E}} = 0$$

$$\gamma = \sqrt{zy} = \alpha + j\beta$$

Lossless Medium

- In a lossless medium:

$$\varepsilon'' = 0 \quad \mu'' = 0 \quad \sigma = 0$$

$$z = j\omega\mu' \quad y = j\omega\varepsilon'$$

$$\gamma = j\omega\sqrt{\mu'\varepsilon'} = j\beta$$

Wavenumber

- For lossless media, it is customary to define the wavenumber (a purely real quantity):

$$k = \omega\sqrt{\mu'\varepsilon'} \qquad \Rightarrow \qquad \beta = k$$

- Helmholtz equation for electric field in terms of wavenumber is

$$\nabla^2\tilde{\vec{E}} + k^2\tilde{\vec{E}} = 0$$

Uniform Plane Wave Solution (1 of 3)

- Let's assume a time-harmonic plane wave traveling in the +z-direction with only an x-component of the electric field.

$$\tilde{\vec{E}} = \hat{x}\tilde{E}_x(z)$$

$$\frac{\partial}{\partial x} \equiv 0$$

$$\frac{\partial}{\partial y} \equiv 0$$

Uniform Plane Wave Solution (2 of 3)

- Helmholtz equation for electric field becomes

$$\frac{\partial^2 \tilde{E}_x}{\partial z^2} + k^2 \tilde{E}_x = 0$$

- The solution is

$$\tilde{E}_x = E_{x0}^{+} e^{-jkz} + E_{x0}^{-} e^{+jkz}$$

Uniform Plane Wave Solution (3 of 3)

- The second term represents a wave propagating in the –z-direction, so we set

$$E_{x0}^{-} = 0$$

- Hence, we have

$$\tilde{\vec{E}} = \hat{x} E_{x0}^{+} e^{-jkz}$$

Question

- Show that the terms in the solution represent forward (traveling in the +z-direction) and backward (traveling in the z-direction) traveling waves?
- Hint: Convert from phasor to time domain and refer to previous lecture.

Uniform Plane Wave Solution (1 of 2)

- The corresponding magnetic field can be found from Faraday's law

$$\boxed{\nabla \times \tilde{\vec{E}} = -j\omega\mu' \tilde{\vec{H}}}$$

- We have

$$\nabla \times \tilde{\vec{E}} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 0 & 0 & -jk \\ E_{x0}^{+} e^{-jkz} & 0 & 0 \end{vmatrix} = -\hat{y}\, jkE_{x0}^{+} e^{-jkz}$$

Uniform Plane Wave Solution (2 of 2)

- Hence

$$\tilde{\vec{H}} = \hat{y}\frac{k}{\omega\mu'}E_{x0}^{+}e^{-jkz} = \hat{y}H_{y0}^{+}e^{-jkz}$$

$$H_{y0}^{+} = \frac{k}{\omega\mu'}E_{x0}^{+} = \frac{\omega\sqrt{\mu'\varepsilon'}}{\omega\mu'}E_{x0}^{+} = \sqrt{\frac{\varepsilon'}{\mu'}}E_{x0}^{+} = \frac{E_{x0}^{+}}{\eta}$$

Intrinsic Impedance of Medium

- Where η is the intrinsic impedance of the medium and has dimensions of ohms.

$$\boxed{\eta = \sqrt{\frac{\mu'}{\varepsilon'}}}$$

- Notice that the intrinsic impedance converts an electric field in units of volts per meter to a magnetic field in units of amps per meter.

Intrinsic Impedance of Free Space

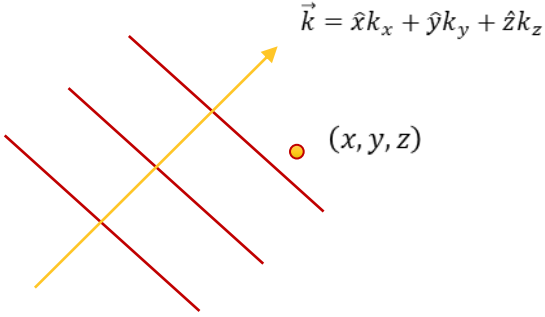
- In free space

$$\boxed{\eta = \sqrt{\frac{\mu_0}{\varepsilon_0}} = \eta_0 = 120\pi(\Omega)}$$

Plane Wave Traveling in Arbitrary Direction in Lossless Simple Medium (1 of 2)

$$\tilde{\vec{E}} = \vec{E}_0 e^{-j\vec{k}\cdot\vec{R}}$$

$$\vec{R} = \hat{x}x + \hat{y}y + \hat{z}z$$



$$\tilde{\vec{E}} = \vec{E}_0 e^{-j\vec{k}\cdot\vec{R}}$$

$$= \vec{E}_0 e^{-jk_x x} e^{-jk_y y} e^{-jk_z z}$$

$$\boxed{\vec{E}_0 \cdot \vec{k} = 0}$$

Plane Wave Traveling in Arbitrary Direction in Lossless Simple Medium (2 of 2)

- Given the electric field the concomitant magnetic field is given by

$$\vec{H} = \frac{1}{\eta}\hat{k} \times \vec{E}$$

- Given the magnetic field, we can compute the concomitant electric field from

$$\vec{E} = -\eta\hat{k} \times \vec{H}$$

These equations are valid for uniform plane waves in simple media (lossy as well as lossless).

TEM Electromagnetic Fields

- Note that

$$\vec{E} \perp \vec{H} \perp \hat{k}$$

- Uniform plane waves are a type of transverse electromagnetic (TEM) wave.
- A TEM wave is one for which the electric field, the magnetic field, and the direction of propagation are mutually perpendicular.

Other TEM Solutions …

- Other TEM solutions to Maxwell's equations exist such as the field in a coaxial cable or other transmission line.
- Such TEM modes can be thought of as a portion of a plane wave limited to the structure and satisfying the boundary conditions of that structure.

Summary

- The propagation constant of a plane wave propagating in a lossless medium is purely imaginary with the phase constant equal to the wavenumber of the medium.
- Plane waves are a type of TEM wave.

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